

GAGANDEEP BHASKAR

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Summary

Master of Applied Artificial Intelligence student at Deakin University with 4+ years of industry experience across AI/ML, Generative AI, Agentic AI, Data Engineering, and Data Analytics. Experienced in delivering production data pipelines, AI solutions, retrieval-augmented and agentic systems, and cloud-based applications across automotive, retail, marketing analytics, and enterprise platforms. Seeking opportunities to apply technical expertise to solve real-world business problems while continuing to grow and contribute to the Australian technology industry.

Experience

Freelancer

Apr 2026 - Present

AI & Data Engineer

- Owning technical direction across parallel client engagements: architecting agentic AI systems with retrieval, tool use, and guardrails; setting up end-to-end observability and evaluation; running production readiness reviews.
- Translating ambiguous business asks into measurable KPI definitions, system architecture, and phased delivery plans; scoping and estimating incoming engagements through structured discovery calls with founders and product leaders.
- Running PR reviews, architecture reviews, and weekly syncs; mentoring engineers on retrieval design, prompt engineering, agent evaluation, and LLM cost/latency trade-offs.

Global Nodes

Sep 2024 - Apr 2026

SDE 2, AI & Data Engineering

- Owned the vision, architecture, and operational delivery of the OOH measurement platform, integrating large-scale mobility data, advanced measurement methodologies, and AI-enabled client intelligence across 130M+ daily records.
- Reformulated the platform's KPI framework (Reach, Frequency, Affinity, and Impressions) from static derivations to multi-factor contextual models, integrating audience behavior, visibility, environmental, and location-based signals to establish the production methodology for client reporting.
- Shipped an agentic anomaly-explanation flow on the platform: for outlier campaign-days, the agent reasons across weather, news, government advisories, and event signals to surface evidence-cited explanations, replacing a manual analyst investigation loop.
- Acted as the AI engineer on additional internal POCs: a trench-measurement computer vision system, an OCR pipeline for the CFPF use case, and a QA automation tool driven by an agentic workflow for test generation and triage.

Unthinkable Solutions LLP

Mar 2022 - Sep 2024

Associate Software Engineer (promoted from Intern)

- Shipped production Python across the ML lifecycle - data prep, training, evaluation, deployment - with delivery spanning object detection and semantic segmentation for structured and unstructured data.
- Evaluated models with precision, recall, F1, and confusion-matrix analysis; mapped model outputs to business KPIs alongside cross-functional product and design teams.
- Led annotation teams and mentored 5+ junior engineers on AI/ML fundamentals and project delivery.

Technical Skills

- **GenAI & Agentic AI:** Agentic RAG, Tool Calling, Model Context Protocol (MCP), Prompt Engineering, Guardrails, LLM Evaluation & Observability, SFT, PEFT/LoRA, Quantization, SLM, Vision LLM and dLLMs
- **GenAI Stack:** LangChain, LangGraph, LangSmith, Langfuse, Hugging Face Transformers, TRL, PEFT, Firecrawl; Vector Stores (pgvector, Pinecone, Chroma, FAISS)
- **AI Coding & Productivity:** Claude Code, Cursor, GitHub Copilot, Claude, ChatGPT
- **Classical AI/ML:** Deep Learning, Computer Vision (Object Detection, Segmentation), Neural Networks; Scikit-learn, TensorFlow, Keras, PyTorch, OpenCV, Ultralytics, Segmentation-Models-PyTorch
- **Data Engineering:** Databricks, Delta Lake, Apache Spark/PySpark, Apache Sedona, AWS Glue, ETL/ELT, CDC, Bronze/Silver/Gold lakehouse design
- **Cloud:** AWS, Databricks
- **Engineering Practice:** Git, GitHub, Bitbucket, CI/CD (GitHub Actions), Jira, code & architecture review, technical mentorship

Key Projects

OOH Advertisement Measurement Platform

- Distributed ETL handling **130M+** GPS and foot-traffic records daily on a Medallion lakehouse with Apache Sedona geospatial joins, metadata-driven lineage, and real-time error alerting that pulled **incident detection from hours to minutes**
- Redefined the platform's reach, affinity, frequency, and impression KPIs into multi-factor models: occlusion-based visibility, time-of-day illumination, weather, event-driven footfall and other contextual signals.
- Layered an agentic explanation flow on top of the warehouse: for any campaign-day with anomalous impressions, the agent retrieves and reasons across weather forecasts, news events, government advisories, and local event signals to produce a grounded, evidence-cited explanation of *why* performance moved.

Maruti Suzuki, Side Airbag Deployment Detection

- Detection model for slow-motion airbag-deployment videos identifying critical events (rip-through, shoulder-level reach, full deployment).
- Built annotation workflow on AWS SageMaker Ground Truth; trained UNet segmentation models across airbag categories and material types to **95%+** accuracy; layered the Segment Anything Model (SAM) as a second-stage refinement that lifted production accuracy past **97%**.
- Replaced the manual frame-by-frame QA loop, saving approximately **40 hours per report cycle** while maintaining the precision required for safety-critical sign-off in a CMMI Level 5 manufacturing environment.

Property Data Pipeline

- Spearheaded end-to-end data management processes, including secure ingestion from S3 buckets, orchestrated ETL workflows on AWS Glue, and automated data transformation tasks, achieving an **80%** optimization in data transformation costs.
- Executed robust CI/CD pipelines using GitHub Actions to streamline deployment of ETL scripts to AWS Glue, ensuring seamless updates & modifications while maintaining data integrity.
- Leveraged AWS services such as Secrets Manager and Lambda with Boto3 to enhance security measures, optimize ETL performance, and provide backend users with efficient job management capabilities.

Maruti Suzuki, Part Failure Microscopic Image Analytics

- Forensic analysis of microscopic images from failed automotive parts to identify fracture types and propagation paths for R&D and quality assurance - work that previously required a trained materials engineer to read each image manually.
- Designed a two-stage pipeline: a DeiT-based classifier identifying material type (PP, PA6, etc.) routing each image to a material-specific UNet segmenter that isolated fracture regions and extracted features like Hackles and Origin; post-processing then traced the fracture path and placed directional arrows automatically.
- Hit **90%+** accuracy across material types (IoU and Dice Loss tuned per stage) and saved approximately **60 hours per report cycle**.

Anomaly Detection, Multi-Tenant Retail Platform

- Anomaly detection pipeline for a multi-tenant retail platform with diverse order patterns, seasonality, and baseline behaviour across tenants - where false-positive control matters as much as recall.
- H2O-based detection model with MLflow experiment tracking; delivered data-drift and data-quality reports to keep performance honest as upstream data shifted across tenants.
- Drove an **85%** operational-efficiency lift through earlier detection and clearer routing of suspect events.

Certifications

- **Technical:** Databricks Certified Associate Data Engineer
- **Language:** Pearson Test of English (PTE) : 83 Overall

Education

Deakin University

Master of Applied Artificial Intelligence (Professional)

2026 - Present

Guru Gobind Singh Indraprastha University

Bachelor of Computer Applications, CGPA: 8.96

2019 - 2022